

MOI GIRLS' ELDORET 2025 PREDICTION



POSSIBLE KCSE EXAM PHYSICS

232/3

PAPER 3

Practical

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education (KCSE)

INSTRUCTIONS TO THE CANDIDATES:

- Write your **name** and **index number** in the spaces provided above
- This paper consists of **two question**.
- Answer **all** questions in section **A** and **B** in the spaces provided.
- All working **must** be clearly shown in the spaces provided.
- Mathematical tables and electronic calculators may be used.

For Examiners' Use Only

SECTION	QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
A	1	18	
B	2	22	
TOTAL		40	

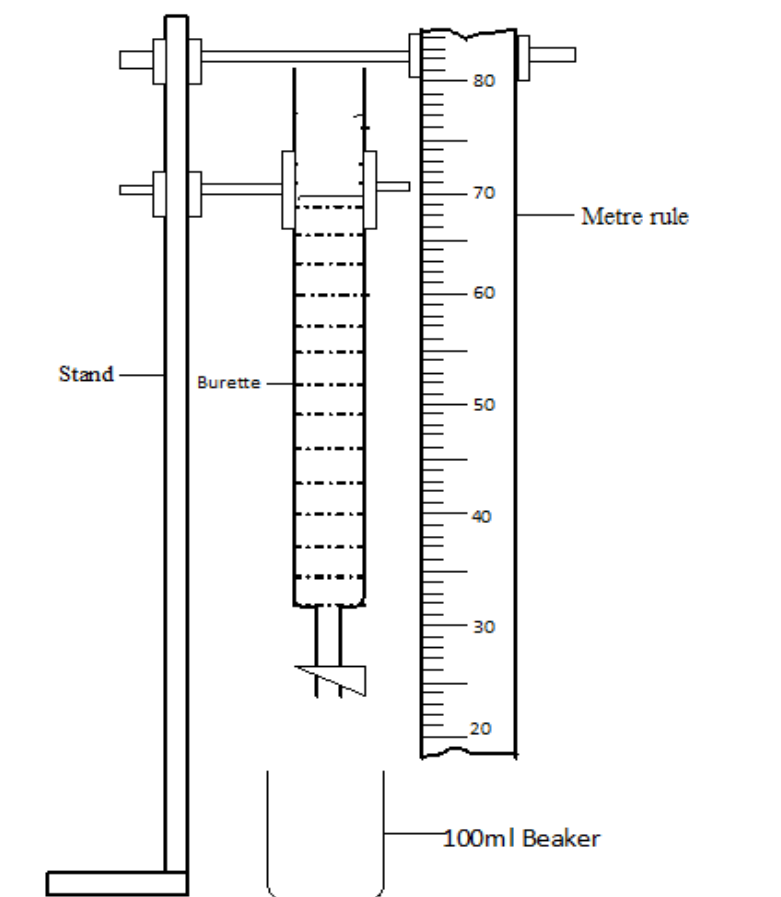
QUESTION ONE

You are provided with the following apparatus

- A clean tap burette
- Retort stand
- Two clamps and 2 bosses
- A metre rule
- Water
- 100ml beaker
- A stop watch

PROCEDURE

- Clamp the burette and metre rule vertically and as close to each other as possible
- Adjust the position of the burette so that its lower end is 15cm above the bench and place the 100ml beaker underneath it.
- Fill the burette with water to a height above the 70cm mark of the metre rule as shown below.



- By trial and error method adjust the rate of flow of the water until the time taken for the water to flow from 70cm mark to 65cm is between 25-30 seconds.
- Once this flow rate has been achieved do not alter the flow rate for the rest of the experiment.
- Fill the burette again with water to a level above the 72cm mark.
- With the water level at the 70cm mark (at $t=0$) start the stop watch. Note the time taken for the height h of the water surface in the burette to decrease by 5cm;Do not stop the watch.
- Continue to record the time taken for the height h of water surface to decrease by successive 5.0cm marks till you have 10 more readings.
- Enter the results in the table below
- Reset the watch and refill the burette .
- Repeat the procedure to get second and third set of readings for t .

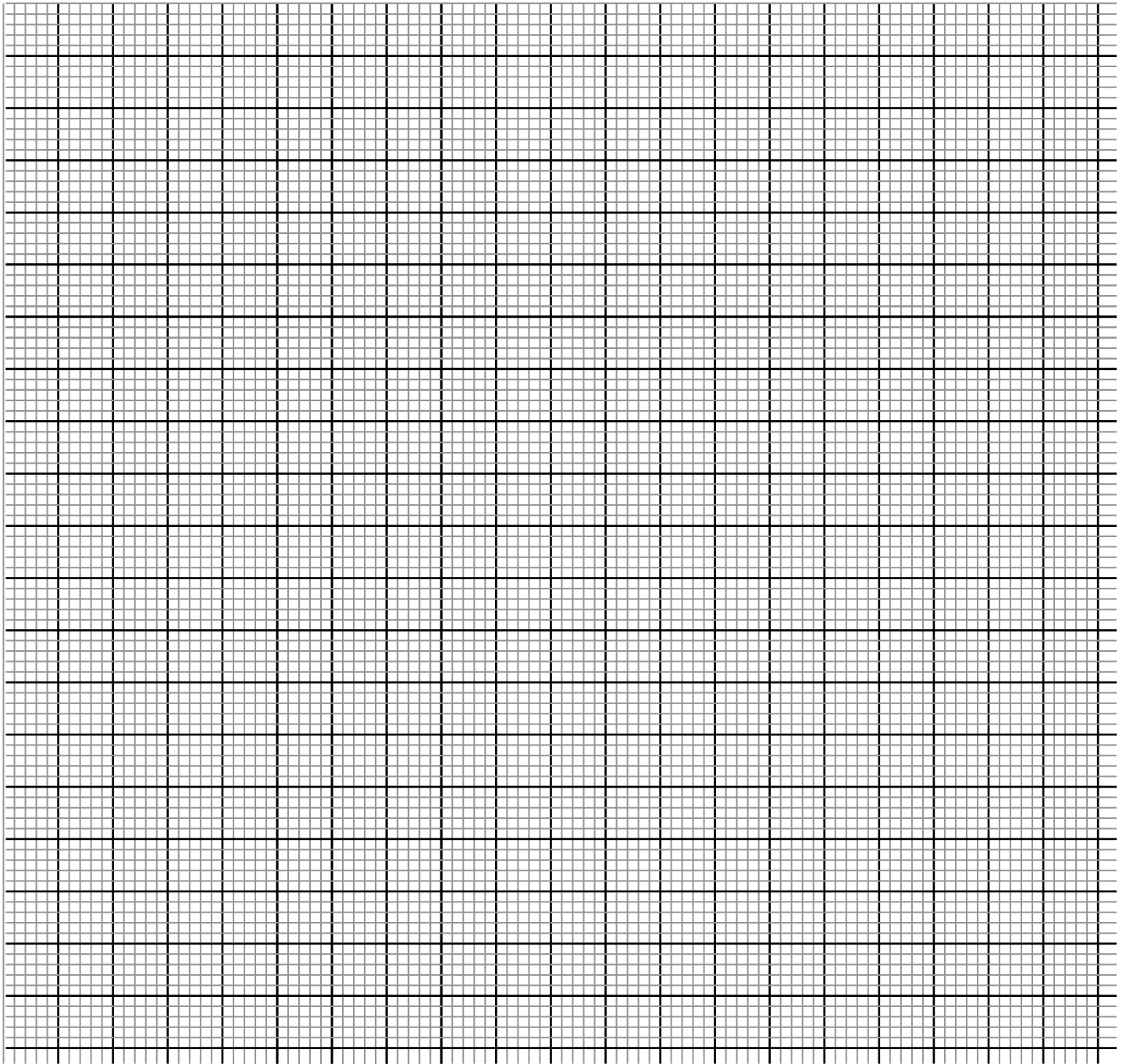
Height h (cm)	Trial 1	Trial 2	Mean time t
70			
65			
60			
55			
50			
45			
40			
35			

(8marks)

Complete the table above and calculate the mean time t

c) Plot a graph of height h (y-axis) against mean time t

(5marks)



d) Use your graph to determine the time taken for the height h to change from 64cm to 42cm. **(2marks)**

e) Find the gradient of the graph when h is 47 cm

3mks

QUESTION 2

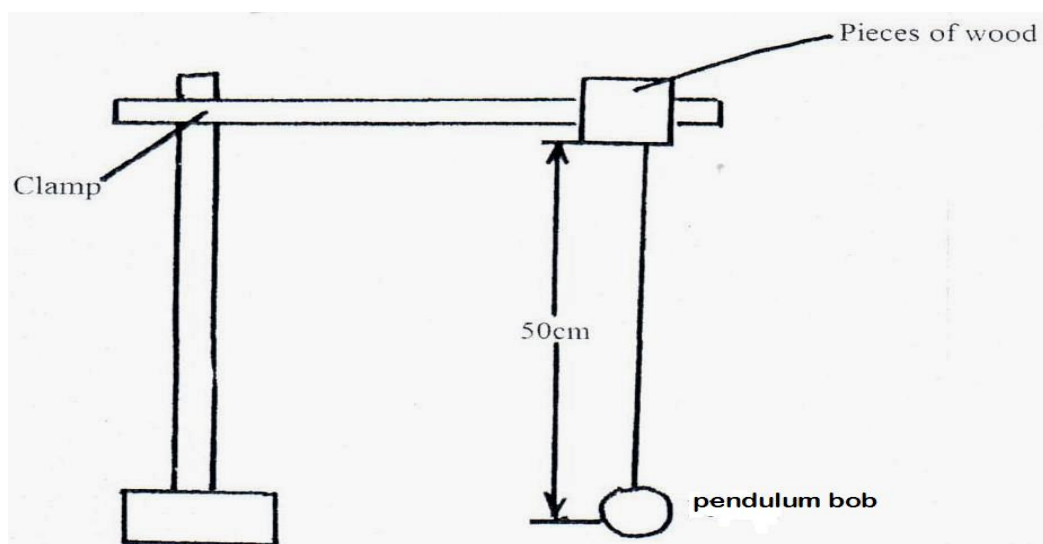
PART A

You are provided with the following:

- Lens
- Lens holder
- Metre rule
- Screen
- A 70cm long thread.
- Stopwatch.
- Clamp, boss and retort stand.
- Small pieces of wood.
- Pendulum bob

Proceed as follows:

- (i) Using the provided thread, tie the marble to be used as a pendulum clamp the thread so that the length l of the pendulum to the centre of the marble is 50cm as in figure 2.



Displace the marble slightly so as to oscillate along the vertical plane.

Time and record the time, t , for 20 oscillations.

$t_1 = \underline{\hspace{2cm}}$ S

(1mk)

$t_2 = \underline{\hspace{2cm}}$ S

(1mk)

$t_{ave} = \underline{\hspace{2cm}}$ S (1mk)

(ii) Calculate the period, $T = \underline{\hspace{2cm}}$ S (2mk)

(iii) Calculate the frequency, $f = \underline{\hspace{2cm}}$ S (2mk)

(iv) $T^2 = \underline{\hspace{2cm}}$ S (1mk)

(v) Convert L to metres, $L = \underline{\hspace{2cm}}$ (1mk)

(v) Calculate $\frac{T^2}{L}$ (1mk)

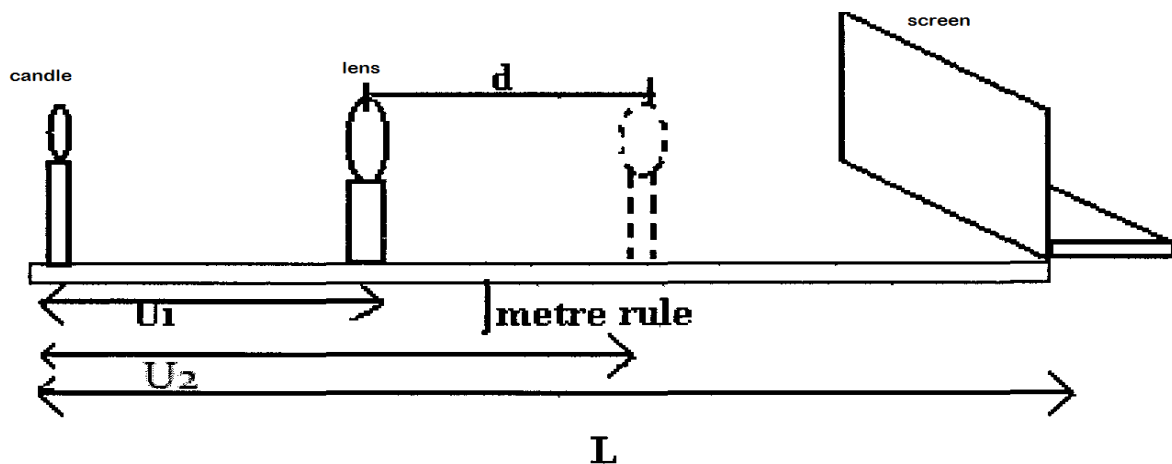
(iv) If the oscillation of the marble is given by the formula

$$\frac{T^2}{L} = \frac{4\pi^2}{g}$$

determine the value of g. (3mks)

PART B

i Set up the apparatus as shown in figure 2 below.(ensure that the candle and the lens are in the line)



ii) With the candle placed a distance $L = 100\text{cm}$ from the screen, determine the position of a magnified image of the candle on the screen by moving the lens

iii) Determine the distance of the lens from the candle U_1

U_1cm (1 mk)

iv) Now move the lens towards the screen until you get a sharply focused diminished image. Determine the new distance of the lens from the candle U_2

U_2cm (1 mk)

v) Calculate the displacement d of the lens (2 mk)

$$d = U_2 - U_1 = \text{.....cm}$$

vi) Give that $\lambda = \frac{L^2 - d^2}{4L}$, calculate the value of λ (3 mks)

vii) By using a distant object, find the focal length f of the lens (1mk)

viii) compare the value in vi and vii (1mk)